

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407487
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Luo; Fa-Long et al.

Full duplex device-to-device cooperative communication

Abstract

Examples described herein include apparatuses and methods for full duplex device-to-device cooperative communication. Example systems described herein may include self-interference noise calculators. The output of a self-interference noise calculator may be used to compensate for the interference experienced due to signals transmitted by another antenna of the same wireless device or system. In implementing such a self-interference noise calculator, a selected wireless relaying device or wireless destination device may operate in a full-duplex mode, such that relayed messages may be transmitted as well as information from other sources or destinations during a common time period (e.g., symbol, slot, subframe, etc.).

Inventors: Luo; Fa-Long (San Jose, CA), Schmitz; Tamara (Scotts Valley, CA), Chritz; Jeremy (Seattle, WA), Cummins; Jaime (Bainbridge Island, WA)

Applicant: MICRON TECHNOLOGY, INC. (Boise, ID)

Family ID: 65631716

Assignee: MICRON TECHNOLOGY, INC. (Boise, ID)

Appl. No.: 17/811472

Filed: July 08, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220353051 A1	Nov. 03, 2022

Related U.S. Application Data

continuation parent-doc US 16718930 20191218 US 11387976 child-doc US 17811472
division parent-doc US 15701007 20170911 US 10554375 20200204 child-doc US 16718930

Publication Classification

Int. Cl.: **H04W72/04** (20230101); **H04B1/525** (20150101); **H04B7/026** (20170101); **H04B7/06** (20060101); **H04B7/08** (20060101); **H04L5/14** (20060101); **H04W4/40** (20180101); **H04W4/70** (20180101); H04L5/00 (20060101); H04L25/02 (20060101); H04W76/14 (20180101)

U.S. Cl.:

CPC **H04L5/14** (20130101); **H04B1/525** (20130101); **H04B7/026** (20130101); **H04B7/0623** (20130101); **H04B7/0854** (20130101); **H04W4/40** (20180201); **H04W4/70** (20180201); H04L5/001 (20130101); H04L25/02 (20130101); H04W76/14 (20180201)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5377108	12/1993	Nishio	N/A	N/A
6199057	12/2000	Tawel	N/A	N/A
6515978	12/2002	Buehrer et al.	N/A	N/A
6765495	12/2003	Dunning et al.	N/A	N/A
7965990	12/2010	Luz et al.	N/A	N/A
9147353	12/2014	Slusar	N/A	N/A
9172441	12/2014	Granger-jones et al.	N/A	N/A
9391680	12/2015	Le-Ngoc et al.	N/A	N/A
9467220	12/2015	Walsh et al.	N/A	N/A
10050663	12/2017	Ku et al.	N/A	N/A
10070432	12/2017	Luo et al.	N/A	N/A
10142137	12/2017	Luo et al.	N/A	N/A
10185998	12/2018	Konrardy et al.	N/A	N/A
10554375	12/2019	Luo et al.	N/A	N/A
10778403	12/2019	Luo et al.	N/A	N/A
10805128	12/2019	Luo et al.	N/A	N/A
10880132	12/2019	Mirfakhraei et al.	N/A	N/A
10890667	12/2020	Chen	N/A	G01S 5/0294
10924139	12/2020	Cong et al.	N/A	N/A
10972139	12/2020	Luo	N/A	N/A
10979097	12/2020	Luo	N/A	N/A
11206050	12/2020	Luo et al.	N/A	N/A
11258473	12/2021	Luo	N/A	N/A
11387976	12/2021	Luo et al.	N/A	N/A
11552658	12/2022	Luo et al.	N/A	N/A
11569851	12/2022	Luo	N/A	N/A
11575548	12/2022	Luo et al.	N/A	N/A

11626966	12/2022	Kolodziej et al.	N/A	N/A
11838046	12/2022	Luo	N/A	N/A
11894957	12/2023	Luo et al.	N/A	N/A
11941516	12/2023	Luo et al.	N/A	N/A
11941518	12/2023	Luo et al.	N/A	N/A
11973525	12/2023	Luo et al.	N/A	N/A
12177167	12/2023	Luo et al.	N/A	N/A
12216234	12/2024	Sumi	N/A	G01S 7/52041
2002/0155821	12/2001	Louis et al.	N/A	N/A
2002/0172265	12/2001	Kenney	N/A	N/A
2005/0060069	12/2004	Breed et al.	N/A	N/A
2005/0078615	12/2004	Muri	N/A	N/A
2006/0162985	12/2005	Tanaka et al.	N/A	N/A
2007/0160014	12/2006	Larsson	N/A	N/A
2008/0117411	12/2007	Vuong et al.	N/A	N/A
2008/0144562	12/2007	Draper et al.	N/A	N/A
2008/0161986	12/2007	Breed	N/A	N/A
2009/0175214	12/2008	Sfar et al.	N/A	N/A
2009/0180404	12/2008	Jung et al.	N/A	N/A
2009/0233555	12/2008	Nakamura	N/A	N/A
2009/0316842	12/2008	Lu et al.	N/A	N/A
2010/0027688	12/2009	Suh et al.	N/A	N/A
2010/0136900	12/2009	Seki	N/A	N/A
2010/0191391	12/2009	Zeng	N/A	N/A
2010/0254300	12/2009	Gulasekaran	370/315	H04B 7/2606
2011/0106442	12/2010	Desai et al.	N/A	N/A
2011/0171922	12/2010	Kim et al.	N/A	N/A
2011/0216662	12/2010	Nie et al.	N/A	N/A
2012/0106405	12/2011	Lioliou et al.	N/A	N/A
2012/0140797	12/2011	Malkin et al.	N/A	N/A
2012/0170619	12/2011	Chang et al.	N/A	N/A
2012/0300680	12/2011	Pietsch et al.	N/A	N/A
2013/0044791	12/2012	Rimini et al.	N/A	N/A
2013/0084868	12/2012	Song et al.	N/A	N/A
2013/0155912	12/2012	Khojastepour et al.	N/A	N/A
2013/0158852	12/2012	Stahlin et al.	N/A	N/A
2014/0003264	12/2013	Shin	N/A	N/A
2014/0056229	12/2013	Li et al.	N/A	N/A
2014/0153494	12/2013	Liu	370/328	H04B 1/126
2014/0169236	12/2013	Choi et al.	N/A	N/A
2014/0219139	12/2013	Choi et al.	N/A	N/A
2015/0018043	12/2014	Taniuchi et al.	N/A	N/A
2015/0043323	12/2014	Choi et al.	N/A	N/A
2015/0055568	12/2014	Jindal et al.	N/A	N/A
2015/0085944	12/2014	Mobasher et al.	N/A	N/A
2015/0139347	12/2014	Murch et al.	N/A	N/A
2015/0140926	12/2014	Fujio et al.	N/A	N/A
2015/0146765	12/2014	Moffatt et al.	N/A	N/A
2015/0244436	12/2014	Le-ngoc et al.	N/A	N/A

2015/0271831	12/2014	Wang et al.	N/A	N/A
2015/0311931	12/2014	Rozental et al.	N/A	N/A
2016/0036504	12/2015	Khojastepour et al.	N/A	N/A
2016/0036582	12/2015	Jana et al.	N/A	N/A
2016/0072531	12/2015	Abrishamkar	455/65	G05B 13/027
2016/0100415	12/2015	Mishra et al.	N/A	N/A
2016/0226535	12/2015	Choi et al.	N/A	N/A
2016/0226653	12/2015	Bharadia et al.	N/A	N/A
2016/0233903	12/2015	Wu et al.	N/A	N/A
2016/0261331	12/2015	Osseiran	N/A	H04W 72/20
2017/0061625	12/2016	Estrada et al.	N/A	N/A
2017/0091953	12/2016	Bleiweiss et al.	N/A	N/A
2017/0104576	12/2016	Liu et al.	N/A	N/A
2017/0131719	12/2016	Micks et al.	N/A	N/A
2017/0150481	12/2016	Gupta et al.	N/A	N/A
2017/0188264	12/2016	Hwang et al.	N/A	N/A
2017/0237547	12/2016	Eltawil et al.	N/A	N/A
2017/0257180	12/2016	Aggarwal et al.	N/A	N/A
2017/0273090	12/2016	Jung et al.	N/A	N/A
2017/0303285	12/2016	Kenney et al.	N/A	N/A
2018/0006690	12/2017	Shepard et al.	N/A	N/A
2018/0006794	12/2017	Lee et al.	N/A	N/A
2018/0026775	12/2017	Chen et al.	N/A	N/A
2018/0048339	12/2017	Wu et al.	N/A	N/A
2018/0062767	12/2017	Vallese et al.	N/A	N/A
2018/0076947	12/2017	Kazakevich et al.	N/A	N/A
2018/0091284	12/2017	Min et al.	N/A	N/A
2018/0123683	12/2017	Wakabayashi et al.	N/A	N/A
2018/0132190	12/2017	Jeon	N/A	H04W 52/242
2018/0152330	12/2017	Chritz et al.	N/A	N/A
2018/0159615	12/2017	Kim	N/A	N/A
2018/0203095	12/2017	Xie	N/A	G01S 7/023
2018/0206176	12/2017	Panteleev et al.	N/A	N/A
2018/0254930	12/2017	Luo et al.	N/A	N/A
2018/0278290	12/2017	Moorti et al.	N/A	N/A
2018/0302048	12/2017	Donoghue et al.	N/A	N/A
2018/0309526	12/2017	Zhang et al.	N/A	N/A
2018/0315157	12/2017	Ould-ahmed-vall et al.	N/A	N/A
2018/0367192	12/2017	O'Shea et al.	N/A	N/A
2019/0007242	12/2018	Luo et al.	N/A	N/A
2019/0020381	12/2018	Toohar et al.	N/A	N/A
2019/0027824	12/2018	Pajona et al.	N/A	N/A
2019/0028260	12/2018	Karlsson et al.	N/A	N/A
2019/0065945	12/2018	Luo et al.	N/A	N/A
2019/0065951	12/2018	Luo et al.	N/A	N/A
2019/0081766	12/2018	Luo et al.	N/A	N/A
2019/0081767	12/2018	Luo et al.	N/A	N/A
2019/0123442	12/2018	Vannucci et al.	N/A	N/A
2019/0229884	12/2018	Xue et al.	N/A	N/A

2019/0238213	12/2018	Sfar et al.	N/A	N/A
2019/0245565	12/2018	Luo et al.	N/A	N/A
2019/0245566	12/2018	Luo et al.	N/A	N/A
2019/0325928	12/2018	Jain et al.	N/A	N/A
2019/0349945	12/2018	Yeh et al.	N/A	N/A
2019/0393948	12/2018	Zhao et al.	N/A	N/A
2020/0127803	12/2019	Luo et al.	N/A	N/A
2020/0196326	12/2019	Li et al.	N/A	N/A
2020/0205156	12/2019	Adjakple et al.	N/A	N/A
2020/0205230	12/2019	Haustein et al.	N/A	N/A
2020/0358177	12/2019	Ge et al.	N/A	N/A
2020/0366538	12/2019	Luo et al.	N/A	N/A
2020/0405204	12/2019	Howard	N/A	N/A
2020/0412518	12/2019	Luo et al.	N/A	N/A
2021/0075464	12/2020	Luo	N/A	N/A
2021/0258038	12/2020	Luo	N/A	N/A
2021/0275050	12/2020	Ren et al.	N/A	N/A
2021/0320678	12/2020	Luo	N/A	N/A
2021/0409132	12/2020	Bouttier	N/A	H04B 17/391
2022/0108157	12/2021	Hunter et al.	N/A	N/A
2022/0116130	12/2021	Elmaghub et al.	N/A	N/A
2022/0140850	12/2021	Luo	N/A	N/A
2022/0217724	12/2021	Pezeshki et al.	N/A	N/A
2023/0073295	12/2022	Luo et al.	N/A	N/A
2023/0113877	12/2022	Luo et al.	N/A	N/A
2023/0179241	12/2022	Luo	N/A	N/A
2023/0268991	12/2022	Wakabayashi	370/327	H04B 7/15528
2024/0256869	12/2023	Luo et al.	N/A	N/A
2024/0267266	12/2023	Luo et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101547021	12/2008	CN	N/A
101563851	12/2008	CN	N/A
102124665	12/2010	CN	N/A
103220089	12/2012	CN	N/A
103780300	12/2013	CN	N/A
104052529	12/2013	CN	N/A
104067585	12/2013	CN	N/A
104468055	12/2014	CN	N/A
104539341	12/2014	CN	N/A
105075157	12/2014	CN	N/A
105594131	12/2015	CN	N/A
105934889	12/2015	CN	N/A
106034097	12/2015	CN	N/A
106134095	12/2015	CN	N/A
106301419	12/2016	CN	N/A
106549897	12/2016	CN	N/A
106559365	12/2016	CN	N/A

106934426	12/2016	CN	N/A
107124245	12/2016	CN	N/A
110024345	12/2018	CN	N/A
110383720	12/2018	CN	N/A
102015015021	12/2015	DE	N/A
1164758	12/2000	EP	N/A
3075095	12/2015	EP	N/A
3200371	12/2016	EP	N/A
2016529837	12/2015	JP	N/A
2016225908	12/2015	JP	N/A
20070116736	12/2006	KR	N/A
20100095650	12/2009	KR	N/A
20130132817	12/2012	KR	N/A
20160090372	12/2015	KR	N/A
20160127099	12/2015	KR	N/A
20160146655	12/2015	KR	N/A
20190091818	12/2018	KR	N/A
2004095625	12/2003	WO	N/A
2010147280	12/2009	WO	N/A
2016111638	12/2015	WO	N/A
2016154219	12/2015	WO	N/A
2017026970	12/2016	WO	N/A
2017069300	12/2016	WO	N/A
2017121487	12/2016	WO	N/A
2018160664	12/2017	WO	N/A
2019046103	12/2018	WO	N/A
2019050980	12/2018	WO	N/A
2019148106	12/2018	WO	N/A
2019156820	12/2018	WO	N/A
2021211560	12/2020	WO	N/A

OTHER PUBLICATIONS

US 11,424,969 B2, 08/2022, Luo et al. (withdrawn) cited by applicant

U.S. Appl. No. 16/718,930 titled “Full Duplex Device-to-Device Cooperation Communication”, filed Dec. 18, 2019, pp. all. cited by applicant

U.S. Appl. No. 17/018,256 titled “Full Duplex Device-to-Device Cooperative Communication” filed Sep. 11, 2020, pp. all. cited by applicant

U.S. Appl. No. 16/561,868 titled “Wireless Devices and Systems Including Examples of Full Duplex Transmission Using Neural Networks or Recurrent Neural Networks” filed Sep. 5, 2019, pp. all. cited by applicant

U.S. Appl. No. 16/848,514 titled “Self Interference Noise Cancellation to Support Multiple Frequency Bands With Neural Networks or Recurrent Neural Networks” filed Apr. 14, 2020, pp. all. cited by applicant

CN Office Action dated Mar. 29, 2021 for CN Application No. 201880058498.3, pp. all. cited by applicant

EESR dated Apr. 15, 2021 for EP Application No. 18854069.4, pp. all. cited by applicant

EN translation of Final Office Action dated Oct. 20, 2021 for KR Appl. No. 10-2020-7007492, pp. all. cited by applicant

English translation of 2nd Office Action for CN Application No. 201980011797.6, mailed Feb. 28, 2022, pp. all. cited by applicant

English translation of KR Office Action dated Apr. 8, 2022 for KR Appl. No. 10-2022-7006932, pp. all. cited by applicant

English translation of Office Action for CN Application No. 201880014872.X, mailed Aug. 6, 2021, pp. all. cited by applicant

English translation of Office Action for KR Application No. 10-2020-7010358, mailed Aug. 10, 2021, pp. all. cited by applicant

English translation of Office Action for KR Application No. 10-2020-7025370, mailed Sep. 16, 2021, pp. all. cited by applicant

Extended European Search Report for EP Application No. 19751566.1, mailed Oct. 13, 2021, pp. all. cited by applicant

International Preliminary Report on Patentability dated Mar. 17, 2022 for PCT Application No. PCT/US2020/045029, pp. all. cited by applicant

International Preliminary Report on Patentability for Application No. PCT/US2018/049598, dated Mar. 17, 2020, pp. all. cited by applicant

International Search Report and Written Opinion dated Dec. 12, 2018 for PCT Application No. PCT/US2018/049598, 11 pages. cited by applicant

U.S. Appl. No. 17/224,962 titled "Wireless Devices and Systems Including Examples of Full Duplex Transmission Using Neural Networks or Recurrent Neural Networks" filed Apr. 7, 2021, pp. all. cited by applicant

U.S. Appl. No. 16/983,797 titled "Wireless Devices and Systems Including Examples of Full Duplex Transmission" filed Aug. 3, 2020, pp. all. cited by applicant

U.S. Appl. No. 15/890,275 entitled "Self Interference Noise Cancellation to Support Multiple Frequency Bands" filed Feb. 6, 2018, pp. all. cited by applicant

U.S. Appl. No. 16/114,923 titled "Cooperative Learning Neural Networks and Systems" filed Aug. 28, 2018, pp. all. cited by applicant

U.S. Appl. No. 16/116,365 titled "Full Duplex Device-to-Device Cooperative Communication" filed Aug. 29, 2018, pp. all. cited by applicant

E. Lee and H. Yeo; "A Study on the Rear-End Collision Warning System by Considering Different Perception-Reaction Time Using Multi-Layer Perception Neural Network"; 2015 IEEE Intelligent Vehicles Symposium (IV); Jun. 28-Jul. 1, 2015; COEX, Seoul, Korea, pp. all. cited by applicant

U.S. Appl. No. 15/693,142, entitled "Cooperative Learning Neural Networks and Systems", filed Aug. 31, 2017, pp. all. cited by applicant

U.S. Appl. No. 15/701,007 entitled "Full Duplex Device-to-Device Cooperative Communication" filed Sep. 11, 2017, pp. all. cited by applicant

U.S. Appl. No. 16/105,915 titled "Wireless Devices and Systems Including Examples of Full Duplex Transmission", filed Aug. 20, 2018, pp. all. cited by applicant

U.S. Appl. No. 16/113,995 titled "Self Interference Noise Cancellation to Support Multiple Frequency Bands"; filed Aug. 27, 2018, pp. all. cited by applicant

U.S. Appl. No. 15/447,731 entitled "Wireless Devices and Systems Including Examples of Full Duplex Transmission", filed Mar. 2, 2017, pp. all. cited by applicant

Anttila, Lauri et al., "Modeling and Efficient Cancellation of Nonlinear Self-Interference in MIMO Full-Duplex Transceivers", IEEE, Oct. 2014, 7 pgs. cited by applicant

Anttila, Lauri et al., "Modeling and Efficient Cancellation of Nonlinear Self-Interference in MIMO Full-Duplex Transceivers", IEEE Globecom Workshops, Dec. 2014, 7 pgs. cited by applicant

Chang, Bao R. et al., "Intelligent Data Fusion System for Predicting Vehicle Collision Warning Using Vision/GPS Sensing"; Expert Systems with Applications 37 (2010) 2439-2450; Mar. 2010; pp. 1-12. cited by applicant

Chang, Bao Rong et al., "Simulation for Implementation of High-Performance Collision Warning System for Motor Vehicle Safety Using Embedded ANFIS Prediction"; 3rd International Conference on Innovative Computing Information and Control (ICICIC '08); IEEE; 2008; pp. all.

cited by applicant

Chen, C. et al., “A rear-end collision prediction scheme based on deep learning in the Internet of Vehicles”; J. Parallel Distribut. Comput.; Elsevier, 2017, pp. all. cited by applicant

Geevarghese, Biju et al., “CDMA Interference Cancellation Techniques Using Neural Networks in Rayleigh Channels”, IEEE: International Conference on Information Communication and Embedded Systems (ICICES), Feb. 2013, 5 pgs. cited by applicant

Kang, M. et al., “Intrusion Detection System Using Deep Neural Network for In-Vehicle Network Security”; Research Article; The Department of Electronics Engineering, Ewha W. University, Seoul, Republic of Korea, Jun. 7, 2016, pp. all. cited by applicant

Kehtarnavaz, et al., A Transportable Neural-Network Approach to Autonomous Vehicle Following, IEEE Transactions on Vehicular Technology, vol. 47, No. 2, May 1998 (Year: 1998), pp. all. cited by applicant

Kehtarnavaz, Nasser , “A Transportable Neural-Network Approach to Autonomous Vehicle Following”, IEEE Transactions on Vehicular Technology, vol. 47, No. 2, May 1998, pp. all. cited by applicant

Kim, Seunghyeon et al., “Transfer Learning for Automated Optical Inspection”; 2017 International Conference on Neural Networks (IJCNN), Jul. 3, 2017; pp. all. cited by applicant

Kristensen, Toftegaard A. et al., “Advanced Machine Learning Techniques for Self-Interference Cancellation in Full-Duplex Radios”, 53rd Asilomar Conference on Signals, Systems, and Computers, Date of Conference: Nov. 3-6, 2019, pp. 1-5. cited by applicant

Sulaiman, Ahmed et al., “Elman Recurrent Neural Network Application in Adaptive Beamforming of Smart Antenna System”, International Journal of Computer Applications (0975-8887), vol. 129, No. 11, Nov. 2015, pp. 38-43. cited by applicant

Tarver, Chance et al., “Neural Network DPD via Backpropagation Through Neural Network Model of the PA”, 53rd Asilomar Conference on Signals, Systems, and Computers, Date of Conference: Nov. 3-6, 2019, pp. 1-5 and figure 4. cited by applicant

Wang, Li “Exploiting Full Duplex for Device-to-Device Communications in Heterogeneous Networks”, IEEE Communications Magazine, May 2015, pp. all. cited by applicant

Young, Chung-Ping et al., “Highway Vehicle Accident Reconstruction Using Cooperative Collision Warning Based Motor Vehicle Event Data Recorder”; IEEE, 2009, pp. all. cited by applicant

Zhang, Zhongshan et al., “Full Duplex Techniquese for 5G Networks: Self-Interference Cancellation, Protocol Design, and Relay Selection”, IEEE Communications Magazine, May 2015, 10 pgs. cited by applicant

Campolo, Claudia , et al., “Full-Duplex Radios for Vehicular Communications”, IEEE Communications Magazine, Jun. 2017; pp. all. cited by applicant

Tran , et al., “On the Performance of Distributed Space-Time Coded Cooperative Relay Networks Based on Inter-Relay Communications”, EURASIP Journal on Wireless Communications and Networking, Oct. 2013; pp. all. cited by applicant

Wang, Li , et al., “Exploiting Full Duplex for Device-to-Device Communications in Heterogeneous Networks”, IEEE Communications Magazine, May 2015; pp. all. cited by applicant

Notice of Preliminary Rejection dated Aug. 18, 2022 for KR application No. 10-2022-7017694; pp. all. cited by applicant

Linsen Chong et al. “A Rule-Based Neural Network Approach to Model Driver Naturalistic Behavior in Traffic”; Transportation Research Part C; www.elsevier.com/locate/trc; vol. 32, Jul. 2013, pp. 207-223. cited by applicant

U.S. Appl. No. 18/065,062, filed Dec. 13, 2022, titled, “Self Interference Noise Cancellation to Support Multiple Frequency Bands,” pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 18/161,600, filed Jan. 30, 2023 titled, “Self Interference Noise Cancellation to Support Multiple Frequency Bands With Neural Networks or Recurrent Neural Networks,” pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/821,419, titled, “Wireless Devices and Systems Including Examples of Full Duplex Transmission,” filed Aug. 22, 2022, pp. all pages of application as filed. cited by applicant
Jinshuan Peng et al. “Multi-Parameter Prediction of Drivers' Lane-Changing Behaviour With Neural Network Model”; Applied Ergonomics, vol. 50, Sep. 2015, pp. 207-217. cited by applicant
U.S. Appl. No. 18/609,221 titled “Cooperative Learning Neural Networks and Systems” filed Mar. 19, 2024, pp. all pages of the application as filed. cited by applicant
U.S. Appl. No. 18/447,611, titled “Wireless Devices and Systems Including Examples of Full Duplex Transmission using Neural Networks or Recurrent Neural Networks” filed Aug. 10, 2023, pp. all pages of application as filed. cited by applicant
“3GPP Release 13 Overview: Summary of all REL-13 Features”, ETSI Mobile Competence Centre, Aug. 2016, 559 pgs. cited by applicant
Feng, Wenjiang , et al. , “Performance Analysis of Distributed Alamouti Space-Time Codes in Full-Duplex Cooperative Communication System” , Journal on Communications, vol. 38, No. 5, May 2017, pp. all. cited by applicant
Liu, Yi, et al. , “Self-Coded Distributed Space-Time Coding for Two-Way Full-Duplex Relay Networks”, 2015 IEEE Global Communications Conference, Feb. 2016, pp. all. cited by applicant
Tang, Xuan-Xuan , et al. , “Survey on Buffer-Aided Relay Selection Schemes in Cooperative Communications” , Journal of Military Communications Technology, vol. 38, No. 1, Mar. 2017, pp. all. cited by applicant
Wang, Jun , et al. , “Performance Analysis of Broadband Self-Interference Cancellation at RF Domain in Co-Frequency Co-Time Full Duplex Systems”, Journal on Communication, vol. 37, No. 9, Sep. 2016, pp. all. cited by applicant
Choi et al., “Millimeter-Wave Vehicular Communication to Support Massive Automotive Sensing”; Automotive Networking and Applications; IEEE Communications Magazine; Dec. 2016; pp. 160-167. cited by applicant

Primary Examiner: Ko; Sithu

Attorney, Agent or Firm: Dorsey & Whitney LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 16/718,930 filed Dec. 19, 2019 and issued as U.S. Pat. No. 11,387,976 on Jul. 12, 2022, which application is a divisional of U.S. patent application Ser. No. 15/701,007 filed Sep. 11, 2017 and issued as U.S. Pat. No. 10,554,375 on Feb. 4, 2020. The aforementioned applications, and issued patents, are incorporated herein by reference, in its entirety, for any purpose.

BACKGROUND

(1) There is an interest in moving wireless communications to “fifth generation” (5G) systems. 5G promises increased speed and ubiquity, but methodologies for processing 5G wireless communications have not yet been set. Example 5G systems may be implemented using multiple-input multiple-output (MIMO) techniques, including “massive MIMO” techniques, in which multiple antennas (more than a certain number, such as 8 in the case of example MIMO systems) are utilized for transmission and/or receipt of wireless communication signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of a wireless communication system arranged in accordance with embodiments of the disclosure.
- (2) FIGS. 2A-2C are diagrams of time-multiplex communication protocols that support full duplex communication in accordance with embodiments of this disclosure.
- (3) FIG. 3 is a schematic illustration of an electronic device arranged in accordance with embodiments of the disclosure.
- (4) FIG. 4 is a schematic illustration of a wireless transmitter in accordance with an embodiment of the disclosure.
- (5) FIG. 5 is a schematic illustration of wireless receiver in accordance with embodiments of this disclosure.
- (6) FIG. 6 is a schematic illustration of an example self-interference noise calculator arranged in accordance with embodiments of the disclosure.
- (7) FIG. 7 is a schematic illustration of a method to implement full duplex communication in device to device cooperative wireless communication in accordance with embodiments of the disclosure.
- (8) FIG. 8 is a schematic illustration of an example of a wireless communications system in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

- (9) Device-to-device cooperative communication may be used in wireless systems that include use of electronic devices to relay messages between a base station and destination electronic device. To improve efficiency, full duplex communication desirable to allow contemporaneous reception and transmission of messages. Full duplex communication generally may refer to an ability to both send and receive transmissions. In examples of systems employing full duplex communication, it may be desirable to cancel interference generated by other antennas in the system. Examples described herein may compensate for interference generated by other antennas co-located on the same physical device or system (e.g., interference created by an antenna on a MIMO device).
- (10) In the example of full duplexing (FD), an antenna transmitting a transmission on a certain frequency band may create interference for an antenna, co-located on the same device, receiving a transmission on the same frequency band. Such interference may be referred to as self-interference. Self-interference may disrupt the accuracy of signals transmitted or received by the MIMO device. Examples described herein may compensate for self-interference at an electronic device, which may aid in achieving full duplex transmission. A network of processing elements may be used to generate adjusted signals to compensate for self-interference generated by the antennas of the electronic device.
- (11) 5G systems may advantageously make improved usage of full duplex transmission mode, for example, to improve spectrum efficiency. Frequency bands in some systems may be assigned by regulatory authorities such as the Federal Communication Commission (FCC) in the United States or similar national- or regional-level authorities in other jurisdictions. Assignments may be made, for example, according to different applications such as digital broadcasting and wireless communication. These licensed and assigned frequencies may be underused if devices simply employ time-division duplex (TDD), frequency-division duplex (FDD) or half-duplex FDD mode, which are duplexing modes often used in existing wireless applications. Such modes may limit the extent to which available spectrum resources may be exploited despite ever-increasing user demands for improved efficiency. Moreover, with the fast development of digital transmission and communications, there is increasing competition for access to unlicensed frequency bands and it may be advantageous to use both licensed and unlicensed frequency bands in a full duplex transmission mode.
- (12) For example, the FCC has officially proposed to open some UHF bands for unlicensed uses

and is also considering how to use the frequency bands which are over 6 GHz (e.g. millimeter wave bands). Examples described herein may be utilized to achieve full duplex transmission in some examples on existing frequency bands including the aforementioned unlicensed frequency bands and 6 GHz bands. Full-duplex (FD) operation described herein may allow a wireless communication system to transmit and receive the signals, contemporaneously, in the same frequency band using spatial diversity. As used herein, contemporaneous means occurring during the same time period. Thus, messages or signals transmitted or received contemporaneously mean transmitted or received during the same time period, such as a single symbol period, slot, subframe, or other discrete duration. This may allow FD 5G systems to the spectrum efficiency of a frequency band.

(13) FIG. 1 is a schematic illustration of a wireless communication system **100** arranged in accordance with embodiments of the disclosure. The system **100** includes a base station **110**, electronic devices **120(0)-120(3)**, and an electronic device **130**. The base station **110** is in communication with the electronic device **130** via the electronic devices **120(0)-120(3)**. That is, the electronic devices **120(0)-120(3)** may implement device-to-device cooperative communication to relay information between the base station **110** and the electronic device **130**. The electronic devices **120(0)-(3)** and/or the electronic device **130** may be referred to as user equipment (UE), mobile terminals, or the like. The electronic devices **120(0)-(3)** or the electronic device **130**, or both, may be a particular category or class of UE, as may be defined by a wireless communication standard (e.g., LTE, 5G New Radio, etc.).

(14) A category or class of UE may indicate or include various characteristics or capabilities, including number of antennas, duplexing capability, spatial multiplexing capability, or the like. The electronic devices **120(0)-(3)** or the electronic device **130**, or both, may provide an indication of their category or capability to base station **110**, or to a network via base station **110**. In some examples, electronic devices **120(0)-(3)** or the electronic device **130** may provide an indication of their respective category or capability to one another. The electronic devices **120(0)-(3)** or the electronic device **130** may implement the techniques described herein or may be requested by one another or by base station **110** to implement techniques described herein according to their respective categories or capabilities. Devices having a same or similar category or capability may be referred to as peer devices and may thus facilitate peer-to-peer communications, which may be in contrast to communication between base station **110** and an electronic device **120(0)-(3)** or electronic device **130**.

(15) In operation, the base station **110** may communicate with the electronic devices **120(0)-120(3)** using a wireless communication protocol. Communication originating at base station **110** and terminating at an electronic device **120(0)-(3)** or electronic device **130** may be referred to as a downlink or forward link communication. Such communication may also be referred to as occurring on the downlink. Communication originating at an electronic device **120(0)-(3)** or electronic device **130** and terminating at base station **110** may be referred to as an uplink or reverse link communication. Such communication may also be referred to as occurring on the uplink. Communication between electronic devices **120(0)-(3)** or between an electronic device **120(0)-(3)** and an electronic device **130** (e.g., D2D communication) may be referred to as a sidelink. Such communication may also be referred to as occurring on the sidelink. The communication may be in accordance with any of a variety of protocols, including, but not limited to long term evolution (LTE), LTE advanced (LTE-A), 5G New Radio (NR) or other standards developed by the 3rd Generation Partnership Project (3GPP), for example. The communication may further include a variety of modulation/demodulation schemes may be used, including, but not limited to: orthogonal frequency division multiplexing (OFDM), filter bank multi-carrier (FBMC), the generalized frequency division multiplexing (GFDM), universal filtered multi-carrier (UFMC) transmission, bi-orthogonal frequency division multiplexing (BFDM), sparse code multiple access (SCMA), non-orthogonal multiple access (NOMA), multi-user shared access (MUSA) and faster-than-Nyquist

(FTN) signaling with time-frequency packing.

(16) The communication between the electronic devices **120(0)-120(3)** and the electronic device **130** may support full duplex communication to relay data from the base station **110** to the electronic device **130** and from the electronic device **130** to the base station **110**. For example, one of the electronic devices **120(0)-120(3)** may relay information from the base station **110** to the electronic device **130**. The relaying communication may be based on techniques that time-division multiplex data into different slots. Examples of time-division multiplex techniques used for relaying data may include generalized selection combining (GSC), distributed space-time coding (DSTC), and opportunistic relaying selection (ORS). Handshake protocols may be implemented to confirm received of data at the electronic device **130** from the base station **110**, or at the base station **110** from the electronic device **130**, such as acknowledgment (ACK) messages.

(17) During a full-duplex communication, the electronic device **130** may contemporaneously receive a message from one of the electronic devices **120(0)-120(3)** and transmit another message (e.g., an ACK message) to the same or another of the electronic devices **120(0)-120(3)** via a common carrier frequency. In some responses, the transmitted message may be in response to a previously received message. In an example of a full-duplex transmission mode, a wireless transmitter and antenna pair of one of the electronic devices **120(0)-120(3)** may communicate with a wireless receiver and antenna pair of the electronic device **130** contemporaneously with a wireless receiver and antenna pair of one (e.g., the same or different one) of the electronic devices **120(0)-120(3)** communicating with a wireless transmitter and antenna pair of the electronic device **130**. The communications may occur during predefined time periods (e.g., symbol periods, slots, subframes, etc.) according to an implemented protocol, such as one of the GSC, DSTC, or ORS protocols.

(18) To enable full-duplex communication, self-interference received by a receiving antenna from a transmitting antenna on the same device (e.g., one of the electronic devices **120(0)-120(3)** or the electronic device **130**) may be compensated for to filter out signals from the transmitted antenna. Self-interference may generally refer to any wireless interference generated by transmissions from antennas of an electronic device to signals received by other antennas, or same antennas, on that same electronic device.

(19) The base station **110**, the electronic devices **120(0)-120(3)**, and the electronic device **130** may be implemented using generally any electronic device for which communication capability is desired. For example, the base station **110** may be implemented using a Wi-Fi access point, an LTE/LTE-A evolved node B (eNB) LTE/LTE-A), or a 5G next generation node B (gNB), or other standardized base station. Each of the electronic devices **120(0)-120(3)** and the electronic device **130** may be implemented using a mobile phone, smartwatch, computer (e.g. server, laptop, tablet, desktop), or radio. In some examples, the electronic devices **120(0)-120(3)** and/or the electronic device **130** may be incorporated into and/or in communication with other apparatuses for which communication capability is desired, such as but not limited to, a wearable device, a medical device, an automobile, airplane, helicopter, appliance, tag, camera, or other device.

(20) While not explicitly shown in FIG. 1, the base station **110**, the electronic devices **120(0)-120(3)** and/or the electronic device **130** may include any of a variety of components in some examples, including, but not limited to, memory, input/output devices, circuitry, processing units (e.g. processing elements and/or processors), or combinations thereof.

(21) Each of the base station **110**, the electronic devices **120(0)-120(3)**, and the electronic device **130** may support multiple input, multiple output (MIMO) systems. MIMO systems generally refer to systems including one or more electronic devices that transmit signals using multiple antennas and one or more electronic devices that receive signals using multiple antennas. In some examples, electronic devices may both transmit and receive signals using multiple antennas. Some example systems described herein may be “massive MIMO” systems. Generally, massive MIMO systems refer to systems employing greater than a certain number (e.g. 64) antennas to transmit and/or

receive transmissions. As the number of antennas increase, so to generally does the complexity involved in accurately transmitting and/or receiving transmissions. Thus, the base station **110**, the electronic devices **120(0)-120(3)**, and the electronic device **130** may each include multiple antennas, including 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 32, or 64 antennas to support a MIMO system. Other numbers of antennas may be used in other examples. In some examples, the electronic devices **120(0)-120(3)** and the electronic device **130** may have a same number of antennas, and a different number of antennas than the base station **110**. In other examples, the base station **110**, the electronic devices **120(0)-120(3)**, and the electronic device **130** may each have different numbers of antennas.

(22) Although one base station **110**, four electronic devices **120(0)-120(3)**, and one electronic device **130** are shown in FIG. 1, the system may include any number of base stations and electronic devices may.

(23) The base station **110**, the electronic devices **120(0)-120(3)**, and the electronic device **130** may each include receivers, transmitters, and/or transceivers. Generally, receivers may be provided for receiving transmissions from one or more connected antennas, transmitters may be provided for transmitting transmissions from one or more connected antennas, and transceivers may be provided for receiving and transmitting transmissions from one or more connected antennas. Generally, multiple receivers, transmitters, and/or transceivers may be provided in an electronic device—one in communication with each of the antennas of the electronic device. The transmissions may be in accordance with any of a variety of protocols, including, but not limited to 5G NR signals, and/or a variety of modulation/demodulation schemes may be used, including, but not limited to: orthogonal frequency division multiplexing (OFDM), filter bank multi-carrier (FBMC), the generalized frequency division multiplexing (GFDM), universal filtered multi-carrier (UFMC) transmission, bi-orthogonal frequency division multiplexing (BFDM), sparse code multiple access (SCMA), non-orthogonal multiple access (NOMA), multi-user shared access (MUSA) and faster-than-Nyquist (FTN) signaling with time-frequency packing. In some examples, the transmissions may be sent, received, or both, in accordance with 5G protocols and/or standards. In some examples, techniques described herein may be employed by a relay node, which may have the same or similar functionality as a base station or access point. Or a relay node may provide more limited functionality than a base station but may be a fixed terminal or node that receives and forwards (e.g., repeats) a signal received from one device (e.g., a UE) to another device (e.g., a base station).

(24) Examples of transmitters, receivers, and/or transceivers may be implemented using a variety of components, including, hardware, software, firmware, or combinations thereof. For example, transceivers, transmitters, or receivers may include circuitry and/or one or more processing units (e.g. processors) and memory encoded with executable instructions for causing the transceiver to perform one or more functions described herein (e.g. software).

(25) FIGS. 2A-2C are diagrams of time-division multiplexed communication techniques that support full duplex communication in accordance with embodiments of this disclosure. FIG. 2A is a diagram **201** of an ORS protocol that supports full duplex communication in accordance with an embodiment of the disclosure. FIG. 2B is a diagram **202** of a GSC protocol that supports full duplex communication in accordance with an embodiment of the disclosure. FIG. 2C is a diagram **203** of a DSTC system that supports full duplex communication in accordance with an embodiment of the disclosure. The base station **210** of FIGS. 2A-2C may be used to implement the base station **110** of FIG. 1. The electronic devices **120(0)-120(3)** of FIGS. 2A-2C may be used to implement the electronic devices **120(0)-120(3)** of FIG. 1. The electronic device **230** of FIGS. 2A-2C may be used to implement the electronic device **130** of FIG. 1.

(26) Referring to FIG. 2A and implementation of an ORS protocol, during a first time period (e.g., Time Slot 1), the base station **210** may transmit a first message MSG1 to the electronic devices **220(0)-220(3)**. In an ORS system, one of the electronic devices **220(0)-220(3)** is selected to relay the MSG1 message from the base station **210** to the electronic device **230** on the basis of various

characteristics, such as signal quality, signal interference, history of successful transmissions, stability of potential relay devices, etc. Therefore, as shown in FIG. 2A, during a second time period (e.g., Time Slot 2), the electronic device **220(1)** is selected to relay the MSG1 message to the electronic device **230**. Contemporaneously with transmission of the MSG1 message by the electronic device **220(1)** (e.g., during Time Slot 2), the electronic device **230** may transmit a second message MSG2 to the electronic devices **220(0)-220(3)**, similar to the manner in which the base station **210** transmits the MSG1 to the electronic devices **220(0)-220(3)**. The MSG2 message may be in response to a previously received message from one of the electronic devices **220(0)-220(3)**. For example, MSG2 may be an acknowledgment message associated a previously-received message (e.g., a message sent from base station **210** or from one of the electronic devices **220(0)-(3)**). In a subsequent time period (not shown), one of the electronic devices **220** may be chosen to transmit the MSG2 to the base station **210**. The manner or criteria for choosing the device **220** for transmission from an electronic device **230** to base station **210** may be similar to the manner or criteria for choosing an electronic device **230** to transmit the MSG1 message during the Time Slot 2. To support the contemporaneous transmission from and reception at the electronic device **230**, the electronic device **230** may include a self-interference calculator to determine interference at a receiving antenna of the electronic device **230** caused by transmission of the MSG2 message to provide an interference signal and a compensation circuit to compensate for that interference in a received signal at the receiving antenna based on the interference signal. By compensating for the interference, the MSG1 message may be successfully demodulated and decoded at the electronic device **230**. It is understood that the contemporaneous transmissions described with reference to FIG. 2A may use a common carrier frequency, in some examples. The transmissions may use different carrier frequencies in other examples.

(27) Referring to FIG. 2B and implementation of a GSC protocol, during a first time period (e.g., Time Slot 1, which may be symbol period, slot, subframe, or the like), the base station **210** may transmit a first message MSG1 to the electronic devices **220(0)-220(3)**. In an a GSC system, the MSG1 message may divided into multiple parts to be transmitted by a selected two or more of the electronic devices **220(0)-220(3)** in a time division duplex (TDD) protocol, with each of the selected two or more of the electronic devices **220(0)-220(3)** transmitting an assigned part of the MSG1 message during an assigned time slot. The associated GSC specification may define an algorithm for segmentation and assignment or selection of the two or more electronic devices **220(0)0(3)**.

(28) In the example depicted in FIG. 2, during a time period S1, the electronic device **220(1)** may transmit a first part of the MSG1 message to the electronic device **230**. During a time period S2, the electronic device **220(2)** may transmit a second part of the MSG1 message to the electronic device **230**. During a time period S3, the electronic device **220(3)** may transmit a third part of the MSG1 message to the electronic device **230**. A number of selected relays may be based on various characteristics, such number of available relays, traffic through each relay, signal quality, magnitude of interference, history of successful transmissions, stability of potential relays, etc. The associated GSC specification may define that the selected relays are identified in a control plan (e.g., or control channel) associated with all of the related devices (e.g., including some or all of the base station **210**, the electronic devices **220(0)-(3)** and the electronic device **230**), and the control plan may use an algorithm defined in the associated GSC specification to perform the selection of the relays.

(29) The number of parts in into which the MSG1 message is divided may be based on a number of selected electronic devices to be used as relays. By dividing the transmission among multiple devices, a communication burden on any single one of the electronic devices **220(0)-220(3)** may be reduced as compared with using a single electronic device to transmit the MSG1 message. In time period 2, the electronic device **220(1)** is selected to relay the MSG1 message to the electronic device **230**. Contemporaneously with transmission of the MSG1 message by the electronic devices

220(1)-220(3) (e.g., during all or part of time slots S1, S2, and S3), the electronic device **230** may transmit a second message MSG2 to the electronic devices **220(0)-220(3)**, similar to the transmission of the MSG1 from the base station **210** to the electronic devices **220(0)-220(3)**. The MSG2 message may be in response to a previously received message from one or more of the electronic devices **220(0)-220(3)**.

(30) In some examples, in a subsequent time period (not shown), a selected subset (or all) of the electronic devices **220(0)-220(3)** may retransmit a part of the MSG2 message to the base station **210**. To support the contemporaneous transmission and reception at the electronic device **230**, the electronic device **230** may include a self-interference calculator to determine interference at receiving antennas of the electronic device **230** caused by transmission of the MSG2 message from the electronic device **230** and a compensation circuit to compensate for that interference in received signals from each of the electronic devices **220(0)-220(3)** at the respective receiving antenna. By compensating for the interference, the parts of the MSG1 message may be successfully demodulated, decoded, and recombined to form the MSG1 at the electronic device **230**. It is understood that the contemporaneous transmissions described with reference to FIG. 2B may use a common carrier frequency, in some examples. The transmissions may use different carrier frequencies in other examples.

(31) Referring to FIG. 2C and implementation of a DSTC protocol, during a first time period (e.g., Time Slot 1), the base station **210** may transmit a first message MSG1 to the electronic devices **220(0)-220(3)**. In a DSTC system, the MSG1 message transmitted as data vector $(X1, X2)$. The data vector $(X1, X2)$ may be used to generate codes $(X1, -X2^*)$ and $(X2, X1^*)$, where the “*” indicates a complex conjugate. Two of the electronic devices **220(0)-220(3)** may be selected to each transmit a respective one of the codes $(X1, -X2^*)$ and $(X2, X1^*)$. The associated DSTC specification may define a protocol and an algorithm for generating and transmitting messages that include one or both of the codes $(X1, -X2^*)$. As shown in FIG. 2, during a second time slot (e.g., Time Slot 2), the electronic device **220(0)** may transmit the code $(X1, -X2^*)$ to the electronic device **230** and the electronic device **220(3)** may transmit code $(X2, X1^*)$ to the electronic device **230**. The selected electronic devices of the electronic devices **220(0)-220(3)** may be based on various characteristics, such number of available relays, traffic through each relay, signal quality, magnitude of interference, history of successful transmissions, stability of potential relays, etc. The associated DSTC specification may define that the selected relays are identified in a control plan (e.g., or control channel) associated with all of the related devices (e.g., including some or all of the base station **210**, the electronic devices **220(0)-(3)** and the electronic device **230**), and the control plan may use an algorithm defined in the associated DSTC specification to perform the selection of the relays.

(32) By dividing the transmission among multiple devices, a communication burden on any single one of the electronic devices **220(0)-220(3)** may be reduced as compared with using a single electronic device to transmit the MSG1 message. Time Slot 2), the electronic device **220(1)** is selected to relay the MSG1 message to the electronic device **230**. Contemporaneously with transmission of the MSG1 message by the electronic devices **220(0)** and **220(3)** (e.g., during Time Slot 2), the electronic device **230** may transmit a second message MSG2 to the electronic devices **220(0)-220(3)**, similar to the transmission of the MSG1 from the base station **210** to the electronic devices **220(0)-220(3)**. The MSG2 message transmitted as data vector $(Y1, Y2)$. The MSG2 message may be in response to a previously received message from one or more of the electronic devices **220(0)-220(3)**. In a subsequent time slot (not shown), a two selected electronic devices of the electronic devices **220(0)-220(3)** may retransmit the codes $(Y1, -Y2^*)$ and $(Y2, Y1^*)$, where the “*” indicates a complex conjugate, generated from the data vector $(Y1, Y2)$. To support the contemporaneous transmission and reception at the electronic device **230**, the electronic device **230** may include a self-interference calculator to determine interference at receiving antennas of the electronic device **230** caused by transmission of the MSG2 message from the electronic device **230**

to provide interference signals and a compensation circuit to compensate for that interference in received signals from each of the electronic devices **220(0)-220(3)** at the respective receiving antenna according to the interference signals. By compensating for the interference, the codes ($X1, -X2^*$) and ($X2, X1^*$) of the MSG1 message may be successfully demodulated, decoded, and recombined to form the MSG1 at the electronic device **230**. It is understood that the contemporaneous transmissions described with reference to FIG. 2C may use a common carrier frequency, in some examples. The transmissions may use different carrier frequencies in other examples.

(33) FIG. 3 is a schematic illustration of an electronic device **310** arranged in accordance with embodiments of the disclosure. The electronic device **310** may also include self-interference noise calculator **340**, compensation component **345**, and compensation component **347**. FIG. 3 is a schematic illustration **300** of an electronic device **310** arranged in accordance with embodiments of the disclosure. The electronic device **310** may also include self-interference noise calculator **340**, compensation component **345**, and compensation component **347**. Self-interference noise calculator **340** and wireless transmitter **311, 313** may be in communication with one another. Each wireless transmitter **311, 313** may be in communication with a respective antenna, such as antenna **301, 303**. Each wireless transmitter **311, 313** receives a respective signal to be transmitted, such as signals to be transmitted. The wireless receivers **315, 317** may process the signals to be transmitted with the operations of a radio-frequency (RF) front-end to generate transmitter output data $x_{\text{sub.1}}(n), x_{\text{sub.2}}(n)$. The wireless transmitter **311, 313** may process the signals to be transmitted as a wireless transmitter, for example. The electronic device **310** may be used to implement any of the base station **110**, the electronic devices **120(0)-120(3)**, or the electronic device **130** of FIG. 1, any of the base station **210**, the electronic devices **220(0)-220(3)**, or the electronic device **230** of FIGS. 2A-2C, or combinations thereof.

(34) Self-interference noise calculator **340** and compensation components **345, 347** may be in communication with one another. Each wireless receiver may be in communication with a respective antenna, such as antenna **305, 307** and a respective compensation component, such as compensation component **345, 347**. In some examples, a wireless transmission received at antennas **305, 307** may be communicated to wireless receiver **315, 317** after compensation of self-interference by the respective compensation component **245, 247**. Each wireless receiver **315, 317** processes the received and compensated wireless transmission to produce a respective processed received signal. In other examples, fewer, additional, and/or different components may be provided.

(35) Examples of self-interference noise calculators described herein may generate and provide adjusted signals (e.g., interference signals) to compensation components. So, for example, the self-interference noise calculator **340** may generate adjusted signals $y_{\text{sub.1}}(n), y_{\text{sub.2}}(n)$ and provide such adjusted signals to the compensation components **345, 347**. The self-interference noise calculator **340** may generate such adjusted signals $y_{\text{sub.1}}(n), y_{\text{sub.2}}(n)$ on the basis of transmitter output data $x_{\text{sub.1}}(n), x_{\text{sub.2}}(n)$. The self-interference noise calculator **340** may be in communication with multiple (e.g. all) of the wireless transmitters of the electronic device **310** and all the respective compensation components coupled to respective wireless receivers, and may provide adjusted signals based on transmitter output data.

(36) It may be desirable in some examples to compensate for the self-interference noise to achieve full duplex transmission. For example, it may be desirable for wireless transmitters **311, 313** of the electronic device **310** to transmit wireless transmission signals at a certain frequency band; and, at the same time or simultaneously, wireless receivers **315, 317** receive wireless transmission signals on that same frequency band. The self-interference noise calculator **340** may determine the self-interference contributed from each wireless transmission based on the transmitter output data to compensate each received wireless transmission with an adjusted signal $y_{\text{sub.1}}(n), y_{\text{sub.2}}(n)$. Particularly as wireless communications move toward 5G standards, efficient use of wireless

spectra may become increasingly important.

(37) Examples of self-interference noise calculators described herein may provide the adjusted signals $y_{\text{sub.1}}(n)$, $y_{\text{sub.2}}(n)$ to receiver(s) and/or transceiver(s). Compensation components **345**, **347** may receive the adjusted signals $y_{\text{sub.1}}(n)$, $y_{\text{sub.2}}(n)$ and compensate for an incoming received wireless transmission from antennas **305**, **307**. For example, the compensation components **345**, **347** may combine the adjusted signals with the incoming received wireless transmission in a manner which compensates for (e.g. reduces) self-interference. In some examples, the compensation components **345**, **347** may subtract the adjusted signals $y_{\text{sub.1}}(n)$, $y_{\text{sub.2}}(n)$ from the received wireless transmission to produce compensated received signals for the respective wireless receivers **315**, **317**. The compensation components **345**, **347** may communicate the compensated received signals to the wireless receivers **315**, **317**. The wireless receivers **315**, **317** may process the compensated received signal with the operations of a radio-frequency (RF) front-end. The wireless receiver may process the compensated received signals as a wireless receiver **500**, for example. While the compensation components **345**, **347** have been described in terms of subtracting an adjusting signal from a received wireless transmission, it can be appreciated that various compensations may be possible, such as adjusted signal that operates as a transfer function compensating the received wireless transmission or an adjusted signal that operates as an optimization vector to multiply the received wireless transmission. Responsive to such compensation, electronic device **310** may transmit and receive wireless communications signals in a full duplex transmission mode.

(38) Examples of self-interference noise calculators described herein, including the self-interference noise calculator **340** of FIG. **3** may be implemented using hardware, software, firmware, or combinations thereof. For example, self-interference noise calculator **340** may be implemented using circuitry and/or one or more processing unit(s) (e.g. processors) and memory encoded with executable instructions for causing the self-interference noise calculator to perform one or more functions described herein.

(39) For clarity, FIG. **3** depicts antennas **301**, **303** coupled to wireless transmitters **311**, **313**, respectively, and antennas **305**, **307** coupled to wireless receivers **315**, **317**, respectively. However, without departing from the scope of the disclosure, it is appreciated that one or both of the antennas **301**, **303** may also be coupled to a wireless receiver (e.g., including functionality described with reference to the wireless receivers **315**, **317**), and/or that one or both of the antennas **305**, **307** may also be coupled to a wireless transmitter (e.g., including functionality described with reference to the wireless transmitters **311**, **313**). It is also appreciated that one or more of the wireless transmitters **311**, **313** and the wireless receivers **315**, **317** may be modified to include both transmitting (e.g., functionality described with reference to the wireless transmitters **311**, **313**) and receiving (e.g., functionality described with reference to the wireless receivers **315**, **317**) capabilities (e.g., a transceiver).

(40) FIG. **4** is a schematic illustration of a wireless transmitter **400** in accordance with an embodiment of the disclosure. The wireless transmitter **400** receives a signal to be transmitted and performs operations of an RF-front end to generate wireless communication signals for transmission via the antenna **403**. The wireless transmitter **400** may be utilized to implement in any wireless transmitter of any of the base station **110**, the electronic devices **120(0)**-**120(3)**, or the electronic device **130** of FIG. **1**, any of the base station **110**, the electronic devices **220(0)**-**220(3)**, or the electronic device **230** of FIG. **2**, or the wireless transmitters **311**, **313** in FIG. **3**, for example. The transmitter output data $x_{\text{sub.N}}(n)$ is amplified by a power amplifier **432** before the output data are transmitted on an RF antenna **436**. The operations of the RF-front end may generally be performed with analog circuitry or processed as a digital baseband operation for implementation of a digital front-end. The operations of the RF-front end include a scrambler **404**, a coder **408**, an interleaver **412**, a modulation mapping **416**, a frame adaptation **420**, an IFFT **424**, a guard interval **428**, and frequency up-conversion **430**.

(41) The scrambler **404** converts the input data to a pseudo-random or random binary sequence. For example, the input data may be a transport layer source (such as MPEG-2 Transport stream and other data) that is converted to a Pseudo Random Binary Sequence (PRBS) with a generator polynomial. While described in the example of a generator polynomial, various scramblers **404** are possible. The coder **408** may encode the data outputted from the scrambler to code the data. For example, a Reed-Solomon (RS) encoder or turbo encoder may be used as outer coder to generate a parity block for each randomized transport packet fed by the scrambler **404**. In some examples, the length of parity block and the transport packet can vary according to various wireless protocols. The interleaver **412** may interleave the parity blocks output by the coder **408**, for example, the interleaver **412** may utilize convolutional byte interleaving. In some examples, additional coding and interleaving can be performed after the coder **408** and interleaver **412**. For example, additional coding may include an inner coder that may further code data output from the interleaver, for example, with a punctured convolutional coding having a certain constraint length. Additional interleaving may include an inner interleaver that forms groups of joined blocks. While described in the context of a RS coding, Turbo coding, and punctured convolution coding, various coders **408** are possible, such as a low-density parity-check (LDPC) coder or a polar coder. While described in the context of convolutional byte interleaving, various interleavers **412** are possible.

(42) The modulation mapping **416** modulates the data outputted from the interleaver **412**. For example, quadrature amplitude modulation (QAM) can map the data by changing (e.g., modulating) the amplitude of the related carriers. Various modulation mappings can be possible, including, but not limited to: Quadrature Phase Shift Keying (QPSK), SCMA NOMA, and MUSA (Multi-user Shared Access). Output from the modulation mapping **416** may be referred to as data symbols. While described in the context of QAM modulation, various modulation mappings **416** are possible. The frame adaptation **420** may arrange the output from the modulation mapping according to bit sequences that represent corresponding modulation symbols, carriers, and frames.

(43) The IFFT **424** may transform symbols that have been framed into sub-carriers (e.g., by frame adaptation **420**) into time-domain symbols. Taking an example of a 5G wireless protocol scheme, the IFFT can be applied as N-point IFFT:

$$(44) \quad x_k = \sum_{n=1}^N X_n e^{i2\pi kn/N} \quad (1)$$

where $X_{\text{sub}.n}$ is the modulated symbol sent in the n'th 5G sub-carrier. Accordingly, the output of the IFFT **424** may form time-domain 5G symbols. In some examples, the IFFT **424** may be replaced by a pulse shaping filter or poly-phase filtering banks to output symbols for frequency up-conversion **430**. The guard interval **428** adds a guard interval to the time-domain 5G symbols. For example, the guard interval may be a fractional length of a symbol duration that is added, to reduce inter-symbol interference, by repeating a portion of the end of a time-domain 5G symbol at the beginning of the frame. For example, the guard interval can be a time period corresponding to the cyclic prefix portion of the 5G wireless protocol scheme. The frequency up-conversion **430** may up-convert the time-domain 5G symbols to a specific radio frequency. For example, the time-domain 5G symbols can be viewed as a baseband frequency range and a local oscillator can mix the frequency at which it oscillates with the 5G symbols to generate 5G symbols at the oscillation frequency. A digital up-converter (DUC) may also be utilized to convert the time-domain 5G symbols. Accordingly, the 5G symbols can be up-converted to a specific radio frequency for an RF transmission. Before transmission, at the antenna **403**, a power amplifier **432** may amplify the transmitter output data $x_{\text{sub}.N(n)}$ **410** to output data for an RF transmission in an RF domain at the antenna **436**. The antenna **436** may be an antenna designed to radiate at a specific radio frequency. For example, the antenna **436** may radiate at the frequency at which the 5G symbols were up-converted. Accordingly, the wireless transmitter **400** may transmit an RF transmission via the antenna **436** according to the signal to be transmitted **411** received at the scrambler **404**. As described above with respect to FIG. 4, the operations of the wireless transmitter **400** can include a

variety of processing operations. Such operations can be implemented in a conventional wireless transmitter, with each operation implemented by specifically-designed hardware for that respective operation. For example, a DSP processing unit may be specifically-designed to implement the IFFT **424**. As can be appreciated, additional operations of wireless transmitter **400** may be included in a conventional wireless receiver.

(45) FIG. **5** is a schematic illustration of wireless receiver **500** in accordance with embodiments of this disclosure. The wireless receiver **500** receives input data X (i,j) from an antenna **504** and performs operations of a RF wireless receiver to generate receiver output data at the descrambler **544**. The wireless receiver **500** may be utilized to implement in any wireless receiver of any of the base station **110**, the electronic devices **120(0)-120(3)**, or the electronic device **130** of FIG. **1**, any wireless receiver of any of the base station **210**, the electronic devices **220(0)-220(3)**, or the electronic device **230** of FIG. **2**, or the wireless receivers **315**, **317** in FIG. **3**. The antenna **504** may be an antenna designed to receive at a specific radio frequency. The operations of the RF wireless receiver may be performed with analog circuitry or processed as a digital baseband operation for implementation of a digital front-end. The operations of the RF wireless receiver include a frequency down-conversion **512**, guard interval removal **516**, a fast Fourier transform (FFT) **520**, synchronization **524**, channel estimation **528**, a demodulation mapping **532**, a de-interleaver **536**, a decoder **540**, and a descrambler **544**.

(46) The frequency down-conversion **512** may down-convert the frequency domain symbols to a baseband processing range. For example, continuing in the example of a 5G implementation, the frequency-domain 5G symbols may be mixed with a local oscillator frequency to generate 5G symbols at a baseband frequency range. A digital down-converter (DDC) may also be utilized to convert the frequency domain symbols. Accordingly, the RF transmission including time-domain 5G symbols may be down-converted to baseband. The guard interval removal **516** may remove a guard interval from the frequency-domain 5G symbols. The FFT **520** may transform the time-domain 5G symbols into frequency-domain 5G symbols. Taking an example of a 5G wireless protocol scheme, the FFT can be applied as N-point FFT:

$$(47) X_n = \sum_{k=1}^N x_k e^{-i2\pi kn/N} \quad (2)$$

where $X_{\text{sub}.n}$ is the modulated symbol sent in the n'th 5G sub-carrier. Accordingly, the output of the FFT **520** may form frequency-domain 5G symbols. In some examples, the FFT **520** may be replaced by poly-phase filtering banks to output symbols for synchronization **524**.

(48) The synchronization **524** may detect pilot symbols in the 5G symbols to synchronize the transmitted data. In some examples of a 5G implementation, pilot symbols may be detected at the beginning of a frame (e.g., in a header) in the time-domain. Such symbols can be used by the wireless receiver **500** for frame synchronization. With the frames synchronized, the 5G symbols proceed to channel estimation **528**. The channel estimation **528** may also use the time-domain pilot symbols and additional frequency-domain pilot symbols to estimate the time or frequency effects (e.g., path loss) to the received signal. For example, a channel may be estimated on the basis of N signals received through N antennas (in addition to the antenna **504**) in a preamble period of each signal. In some examples, the channel estimation **528** may also use the guard interval that was removed at the guard interval removal **516**. With the channel estimate processing, the channel estimation **528** may compensate for the frequency-domain 5G symbols by some factor to minimize the effects of the estimated channel. While channel estimation has been described in terms of time-domain pilot symbols and frequency-domain pilot symbols, other channel estimation techniques or systems are possible, such as a MIMO-based channel estimation system or a frequency-domain equalization system. The demodulation mapping **532** may demodulate the data outputted from the channel estimation **528**. For example, a quadrature amplitude modulation (QAM) demodulator can map the data by changing (e.g., modulating) the amplitude of the related carriers. Any modulation mapping described herein can have a corresponding demodulation mapping as performed by

demodulation mapping 532. In some examples, the demodulation mapping 532 may detect the phase of the carrier signal to facilitate the demodulation of the 5G symbols. The demodulation mapping 532 may generate bit data from the 5G symbols to be further processed by the de-interleaver 536.

(49) The de-interleaver 536 may deinterleave the data bits, arranged as parity block from demodulation mapping into a bit stream for the decoder 540, for example, the de-interleaver 536 may perform an inverse operation to convolutional byte interleaving. The de-interleaver 536 may also use the channel estimation to compensate for channel effects to the parity blocks. The decoder 540 may decode the data outputted from the scrambler to code the data. For example, a Reed-Solomon (RS) decoder or turbo decoder may be used as a decoder to generate a decoded bit stream for the descrambler 544. For example, a turbo decoder may implement a parallel concatenated decoding scheme. In some examples, additional decoding deinterleaving may be performed after the decoder 540 and de-interleaver 536. For example, additional coding may include an outer coder that may further decode data output from the decoder 540. While described in the context of a RS decoding and turbo decoding, various decoders 540 are possible, such as low-density parity-check (LDPC) decoder or a polar decoder. The descrambler 544 may convert the output data from decoder 540 from a pseudo-random or random binary sequence to original source data. For example, the descrambler 544 may convert decoded data to a transport layer destination (e.g., MPEG-2 transport stream) that is descrambled with an inverse to the generator polynomial of the scrambler 404 of FIG. 4. The descrambler thus outputs receiver output data. Accordingly, the wireless receiver 500 receives an RF transmission including input data X (i,j) 410 via to generate the receiver output data.

(50) As described above with respect to FIG. 5, the operations of the wireless receiver 500 can include a variety of processing operations. Such operations can be implemented in a conventional wireless receiver, with each operation implemented by specifically-designed hardware for that respective operation. For example, a DSP processing unit may be specifically-designed to implement the FFT 520. As can be appreciated, additional operations of wireless receiver 500 may be included in a conventional wireless receiver.

(51) Further, in some embodiments, the wireless transmitter 400 of FIG. 4 and the wireless receiver 500 of FIG. 5 may be combined into a single wireless transceiver capable of performing both transmitting and receiving functions described with reference to FIGS. 4 and 5, respectively.

(52) FIG. 6 is a schematic illustration of an example self-interference noise calculator 600 arranged in accordance with embodiments of the disclosure. The self-interference noise calculator 600 may be utilized to implement the self-interference noise calculator 340 of FIG. 3, for example. The self-interference noise calculator 600 includes a network of processing elements 615, 625, 635 that output adjusted signals $y_{\text{sub}.1}(n)$, $y_{\text{sub}.2}(n)$, $y_{\text{sub}.3}(n)$, $y_{\text{sub}.L}(n)$ on the basis of transmitter output data $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$. For example, the transmitter output data $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$ may correspond to inputs for respective antennas of each transmitter generating the respective $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$. The processing elements 615 receive the transmitter output data $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$ as inputs. The processing elements 615 may be implemented, for example, using bit manipulation units that may forward the transmitter output data $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$ to processing elements 625. Processing elements 625 may be implemented, for example, using multiplication units that include a non-linear vector set (e.g., center vectors) based on a non-linear function, such as a Gaussian function

(53) (e.g., $f(r) = \exp(-\frac{r^2}{2})$,

a multi-quadratic function (e.g., $f(r) = (r_{\text{sup}.2} + \sigma_{\text{sup}.2})$), an inverse multi-quadratic function (e.g., $f(r) = (r_{\text{sup}.2} + \sigma_{\text{sup}.2})^{-1}$), a thin-plate spline function (e.g., $f(r) = r_{\text{sup}.2} \log(r)$), a piece-wise linear function (e.g., $f(r) = \frac{1}{2}(|r+1| - |r-1|)$), or a cubic approximation function (e.g., $f(r) = \frac{1}{2}(|r_{\text{sup}.3} + 1| - |r_{\text{sup}.3} - 1|)$). In some examples, the parameter a is a real parameter (e.g., a scaling parameter) and

r is the distance between the input signal (e.g., $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$) and a vector of the non-linear vector set. Processing elements **635** may be implemented, for example, using accumulation units that sum the intermediate processing results received from each of the processing elements **625**. In communicating the intermediate processing results, each intermediate processing result may be weighted with a weight 'W'. For example, the multiplication processing units may weigh the intermediate processing results based on a minimized error for the all or some of the adjustment signals that may generated by a self-interference noise calculator.

(54) The processing elements **625** include a non-linear vector set may be denoted as $C_{\text{sub}.i}$ (for $i=1, 2, \dots, H$). H may represent the number of processing elements **625**. With the transmitter output data $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$ received as inputs to processing elements **625**, after forwarding by processing elements **615**, the output of the processing elements **625**, operating as multiplication processing units, may be expressed as $h_{\text{sub}.i}(n)$, such that.

$$h_{\text{sub}.i}(n) = f_{\text{sub}.i}(\|X(n) - C_{\text{sub}.i}\|) \quad (i=1, 2, \dots, H) \quad (3)$$

$f_{\text{sub}.i}$ may represent a non-linear function that is applied to the magnitude of the difference between $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$ and the center vectors $C_{\text{sub}.i}$. The output $h_{\text{sub}.i}(n)$ may represent a non-linear function such as a Gaussian function, multi-quadratic function, an inverse multi-quadratic function, a thin-plate spine function, or a cubic approximation function.

(55) The output $h_{\text{sub}.i}(n)$ of the processing elements **625** may be weighted with a weight matrix 'W'. The output $h_{\text{sub}.i}(n)$ of the processing elements **625** can be referred to as intermediate processing results of the self-interference noise calculator **600**. For example, the connection between the processing elements **625** and processing elements **635** may be a linear function such that the summation of a weighted output $h_{\text{sub}.i}(n)$ such that the adjusted signals $y_{\text{sub}.1}(n)$, $y_{\text{sub}.2}(n)$, $y_{\text{sub}.3}(n)$, $y_{\text{sub}.L}(n)$ may be expressed, in Equation 4 as:

$$y_{\text{sub}.1}(n) = \sum_{j=1}^{\text{sup}.H} W_{\text{sub}.ij} h_{\text{sub}.j}(n) = \sum_{j=1}^{\text{sup}.H} W_{\text{sub}.ij} f_{\text{sub}.j}(\|X(n) - C_{\text{sub}.j}\|) \quad (i=1, 2, \dots, L) \quad (4)$$

Accordingly, the adjusted signals $y_{\text{sub}.1}(n)$, $y_{\text{sub}.2}(n)$, $y_{\text{sub}.3}(n)$, $y_{\text{sub}.L}(n)$ **530** may be the output $y_{\text{sub}.i}(n)$ of the i'th processing element **635** at time n, where L is the number of processing elements **635**. $W_{\text{sub}.ij}$ is the connection weight between j'th processing element **625** and i'th processing element **635** in the output layer. The center vectors $C_{\text{sub}.i}$ and the connection weights $W_{\text{sub}.ij}$ of each layer of processing elements may be determined by a training unit that utilizes sample vectors to train the self-interference noise calculator **600**. Advantageously, the adjusted signals $y_{\text{sub}.1}(n)$, $y_{\text{sub}.2}(n)$, $y_{\text{sub}.3}(n)$, $y_{\text{sub}.L}(n)$ generated from the transmitter output data $x_{\text{sub}.i}(n)$, $x_{\text{sub}.2}(n)$, $x_{\text{sub}.3}(n)$, $x_{\text{sub}.N}(n)$ may be computed with near-zero latency such that self-interference compensation may be achieved in any electronic device including a self-interference noise calculator, such as the self-interference noise calculator **600**. A wireless device or system that implements a self-interference noise calculator **600** may achieve full duplex transmission. For example, the adjusted signals generated by the self-interference noise calculator **600** may compensate-interference that an antenna of the wireless device or system will experience due to signals to be transmitted by another antenna of the wireless device or system.

(56) While the self-interference noise calculator **600** has been described with respect to a single layer of processing elements **625** that include multiplication units, it can be appreciated that additional layers of processing elements with multiplication units may be added between the processing elements **615** and the processing elements **635**. The self-interference noise calculator is scalable in hardware form, with additional multiplication units being added to accommodate additional layers. Using the methods and systems described herein, additional layer(s) of processing elements including multiplication processing units and the processing elements **625** may be optimized to determine the center vectors $C_{\text{sub}.i}$ and the connection weights $W_{\text{sub}.ij}$ of each layer of processing elements including multiplication units.

(57) The self-interference noise calculator **600** can be implemented using one or more processors,

for example, having any number of cores. An example processor core can include an arithmetic logic unit (ALU), a bit manipulation unit, a multiplication unit, an accumulation unit, an adder unit, a look-up table unit, a memory look-up unit, or any combination thereof. In some examples, the self-interference noise calculator **600** may include circuitry, including custom circuitry, and/or firmware for performing functions described herein. For example, circuitry can include multiplication unit, accumulation units, and/or bit manipulation units for performing the described functions, as described herein. The self-interference noise calculator **600** may be implemented in any type of processor architecture including but not limited to a microprocessor or a digital signal processor (DSP), or any combination thereof.

(58) FIG. 7 is a schematic illustration of a method **700** to implement full duplex communication in device to device cooperative wireless communication in accordance with embodiments of the disclosure. The method **700** may be implemented using any of the base station **110**, the electronic devices **120(0)-120(3)**, or the electronic device **130** of FIG. 1, the base station **210**, the electronic devices **220(0)-220(3)**, or the electronic device **230** of FIG. 2, or the electronic device **310** of FIG. 3, or any system or combination of the systems depicted in FIGS. 1-3 described herein. The operations described in steps **710-730** may also be stored as computer-executable instructions in a computer-readable medium.

(59) The method **700** may include transmitting, from a first set of antennas of a first device to a second device, a first message on a carrier during a first time period, at **710**. The first device may include the electronic device **130** of FIG. 1, the electronic device **230** of FIG. 2, or the electronic device **310** of FIG. 3. The first message may be transmitted by a wireless transmitter and antenna pair such as the wireless transmitter **311**/antenna **301** pair or the wireless transmitter **313**/antenna **303** pairs of FIG. 3. Transmission of the first message may include transmitting the first message on a sidelink to the second device.

(60) The first device may include the electronic device **130** of FIG. 1, the electronic device **230** of FIG. 2, or the electronic device **310** of FIG. 3.

(61) The method **700** may further include generating a self-interference noise signal at the first device based at least in part on the first message, at **720**. The self-interference noise signal may be generated by a self-interference noise calculator, such as the self-interference noise calculator **340** of FIG. 3 or the self-interference noise calculator **600** of FIG. 6. The application of the self-interference noise signal may be performed by a compensation component, such as one of the compensation components **345**, **347** of FIG. 3. In some examples, the second message is received from a second relay device, or it may be received from the same relay device to which the first message is transmitted.

(62) The method **700** may further include receiving, from the second device at a second set of antennas of the first device, a second message on the carrier during the first time period, at **730**. The second message may include at least a portion of a message from a third device and is received based at least in part on the self-interference noise signal. The second device may include the electronic devices **120(0)-(3)** of FIG. 1, the electronic devices **220(0)-(3)** of FIG. 2, or the electronic device **310** of FIG. 3. Specifically, the second message may be received by a wireless receiver and antenna pair such as the wireless receiver **315**/antenna **305** pair or the wireless receiver **317**/antenna **307** pairs of FIG. 3. In some examples, the first device includes first user equipment (UE) and the second device includes second UE, the first UE and the second UE may be included in a same category of UE. In some examples, the second device comprises a relay node. Receipt of the second message may include receiving the second message on the sidelink from the second device.

(63) In some examples, the method **700** may further include receiving a third message on the carrier during a second time period that is before the first time period, wherein the first message is transmitted in response to the third message. In this example, the first message may include an acknowledgement (ACK) or a negative acknowledgment (NACK) for the third message. In some

examples, the third device includes a third UE. In other examples, at least one of the second device or the third device comprises a base station, such as the base station **110** of FIG. **1**. In some examples, the first device, the second device, and/or the third device are configured to implement one of a generalized selection combining protocol, a distributed space-time coding protocol, or an opportunistic relaying system protocol.

(64) In some examples the method **700** may further receive a second receive signal. The method **700** may further include generating a second self-interference noise signal based on the first message, and applying the second self-interference noise signal to the second receive signal to filter out interference caused by transmission of the first message to provide a third message. The method **700** may further include combining the second message and third message to form a whole message (e.g., such as described in FIGS. **2B** and/or **2C**). For example, the first received signal may be received during a first portion of the time slot and the second receive signal may be received during a second portion of the time slot. In some examples, the first portion of the time slot is non-overlapping with the second portion of the time slot, as in shown in FIG. **2B**.

(65) The steps **710-740** included in the described example method **700** are for illustration purposes. In some embodiments, these steps may be performed in a different order. In some other embodiments, various blocks may be eliminated. In still other embodiments, various blocks may be divided into additional blocks, supplemented with other blocks, or combined together into fewer blocks. Other variations of these specific blocks are contemplated, including changes in the order of the blocks, changes in the content of the blocks being split or combined into other blocks, etc.

(66) FIG. **8** illustrates an example of a wireless communications system **800** in accordance with aspects of the present disclosure. The wireless communications system **800** includes a base station **810**, a mobile device **815**, a drone **817**, a small cell **830**, and vehicles **840**, **845**. The base station **810** and small cell **830** may be connected to a network that provides access to the Internet and traditional communication links. The base station **810** and/or small cell **830** may be implemented using base stations described herein, such as base station **110** of FIG. **1**, base station **210** of FIG. **2**, and/or electronic device **310** of FIG. **3**. The base station **810**, small cell **830**, mobile device **815**, drone **817**, and/or vehicles **840**, **845** may incorporate a transmitter and/or receiver described herein, such as the wireless transmitter **400** of FIG. **4** and/or receiver **500** of FIG. **5**. The mobile device **815**, drone **817**, and/or vehicles **840**, **845** may be implemented using mobile devices described herein, such as electronic devices **120(0)-120(3)** and/or electronic device **130** of FIG. **1**, electronic devices **220(0)-220(3)** and/or electronic device **230** of FIG. **2**, and/or electronic device **310** of FIG. **3**. The base station **810**, small cell **830**, mobile device **815**, drone **817**, and/or vehicles **840**, **845** may utilize self-interference cancellation techniques described herein. The devices and/or base stations of FIG. **8** may further utilize full-duplex communication as described herein, where the devices and/or base stations are capable of contemporaneously transmitting signals to another device or base station and receiving signals from another device or base station.

(67) The base station **810**, small cell **830**, mobile device **815**, drone **817**, and/or vehicles **840**, **845** may further utilize protocols described herein to relay signals while using full-duplex communication. For example, signals to and from the vehicle **840** may be relayed between one or both of the communication devices **820** or the small cell **830** via the vehicle **845**. In another example, signals to and from the vehicle **845** may be relayed between one or both of the mobile device **815** via the communication device **820**. In another example, signals to and from the small cell **830** may be relayed to the base station **810** via the drone **817**. In some examples, signals to and from the vehicles **840**, **845** may be relayed to the base station **810** or the small cell **830** via the drone **817**. The drone **817** may move to follow traffic to help with communication congestion during high volume traffic.

(68) The system **800** may facilitate a wide-range of wireless communications connections in a 5G system that may include various frequency bands, including but not limited to: a sub-6 GHz band (e.g., 700 MHz communication frequency), mid-range communication bands (e.g., 2.4 GHz), and

mmWave bands (e.g., 24 GHz). Additionally or alternatively, the wireless communications connections may support various modulation schemes, including but not limited to: filter bank multi-carrier (FBMC), the generalized frequency division multiplexing (GFDM), universal filtered multi-carrier (UFMC) transmission, bi-orthogonal frequency division multiplexing (BFDM), sparse code multiple access (SCMA), non-orthogonal multiple access (NOMA), multi-user shared access (MUSA), and faster-than-Nyquist (FTN) signaling with time-frequency packing. Such frequency bands and modulation techniques may be a part of a standards framework, such as Long Term Evolution (LTE) or other technical specification published by an organization like 3GPP or IEEE, which may include various specifications for subcarrier frequency ranges, a number of subcarriers, sidelink/uplink/downlink transmission speeds, TDD/FDD, and/or other aspects of wireless communication protocols.

(69) The system **800** may depict aspects of a radio access network (RAN), and system **800** may be in communication with or include a core network (not shown). The core network may include one or more serving gateways, mobility management entities, home subscriber servers, and packet data gateways. The core network may facilitate user and control plane links to mobile devices via the RAN, and it may be an interface to an external network (e.g., the Internet). Base stations **810**, communication devices **820**, and small cells **830** may be coupled with the core network or with one another, or both, via wired or wireless backhaul links (e.g., S1 interface, X2 interface, etc.).

(70) The system **800** may provide communication links connected to devices or “things,” such as sensor devices, e.g., solar cells **837**, to provide an Internet of Things (“IoT”) framework. Connected things within the IoT may operate within frequency bands licensed to and controlled by cellular network service providers, or such devices or things may. Such frequency bands and operation may be referred to as narrowband IoT (NB-IoT) because the frequency bands allocated for IoT operation may be small or narrow relative to the overall system bandwidth. Frequency bands allocated for NB-IoT may have bandwidths of 1, 5, 10, or 20 MHz, for example.

(71) Additionally or alternatively, the IoT may include devices or things operating at different frequencies than traditional cellular technology to facilitate use of the wireless spectrum. For example, an IoT framework may allow multiple devices in system **800** to operate at a sub-6 GHz band or other industrial, scientific, and medical (ISM) radio bands where devices may operate on a shared spectrum for unlicensed uses. The sub-6 GHz band may also be characterized as and may also be characterized as an NB-IoT band. For example, in operating at low frequency ranges, devices providing sensor data for “things,” such as solar cells **837**, may utilize less energy, resulting in power-efficiency and may utilize less complex signaling frameworks, such that devices may transmit asynchronously on that sub-6 GHz band. The sub-6 GHz band may support a wide variety of uses case, including the communication of sensor data from various sensors devices. Examples of sensor devices include sensors for detecting energy, heat, light, vibration, biological signals (e.g., pulse, EEG, EKG, heart rate, respiratory rate, blood pressure), distance, speed, acceleration, or combinations thereof. Sensor devices may be deployed on buildings, individuals, and/or in other locations in the environment. The sensor devices may communicate with one another and with computing systems which may aggregate and/or analyze the data provided from one or multiple sensor devices in the environment.

(72) In such a 5G framework, devices may perform functionalities performed by base stations in other mobile networks (e.g., UMTS or LTE), such as forming a connection or managing mobility operations between nodes (e.g., handoff or reselection). For example, mobile device **815** may receive sensor data from the user utilizing the mobile device **815**, such as blood pressure data, and may transmit that sensor data on a narrowband IoT frequency band to base station **810**. In such an example, some parameters for the determination by the mobile device **815** may include availability of licensed spectrum, availability of unlicensed spectrum, and/or time-sensitive nature of sensor data. Continuing in the example, mobile device **815** may transmit the blood pressure data because a narrowband IoT band is available and can transmit the sensor data quickly, identifying a time-

sensitive component to the blood pressure (e.g., if the blood pressure measurement is dangerously high or low, such as systolic blood pressure is three standard deviations from norm).

(73) Additionally or alternatively, mobile device **815** may form device-to-device (D2D) connections with other mobile devices or other elements of the system **800**. For example, the mobile device **815** may form RFID, WiFi, MultiFire, Bluetooth, or Zigbee connections with other devices, including communication device **820** or vehicle **845**. In some examples, D2D connections may be made using licensed spectrum bands, and such connections may be managed by a cellular network or service provider. Accordingly, while the above example was described in the context of narrowband IoT, it can be appreciated that other device-to-device connections may be utilized by mobile device **815** to provide information (e.g., sensor data) collected on different frequency bands than a frequency band determined by mobile device **815** for transmission of that information.

(74) Moreover, some communication devices may facilitate ad-hoc networks, for example, a network being formed with communication devices **820** attached to stationary objects (e.g., lampposts in FIG. 8) and the vehicles **840**, **845**, without a traditional connection to a base station **810** and/or a core network necessarily being formed. Other stationary objects may be used to support communication devices **820**, such as, but not limited to, trees, plants, posts, buildings, blimps, dirigibles, balloons, street signs, mailboxes, or combinations thereof. In such a system **800**, communication devices **820** and small cell **830** (e.g., a picocell, femtocell, WLAN access point, cellular hotspot, etc.) may be mounted upon or adhered to another structure, such as lampposts and buildings to facilitate the formation of ad-hoc networks and other IoT-based networks. Such networks may operate at different frequency bands than existing technologies, such as mobile device **815** communicating with base station **810** on a cellular communication band.

(75) The communication devices **820** may form wireless networks, operating in either a hierarchical or ad-hoc network fashion, depending, in part, on the connection to another element of the system **800**. For example, the communication devices **820** may utilize a 700 MHz communication frequency to form a connection with the mobile device **815** in an unlicensed spectrum, while utilizing a licensed spectrum communication frequency to form another connection with the vehicle **845**. Communication devices **820** may communicate with vehicle **845** on a licensed spectrum to provide direct access for time-sensitive data, for example, data for an autonomous driving capability of the vehicle **845** on a 5.9 GHz band of Dedicated Short Range Communications (DSRC).

(76) Vehicles **840** and **845** may form an ad-hoc network at a different frequency band than the connection between the communication device **820** and the vehicle **845**. For example, for a high bandwidth connection to provide time-sensitive data between vehicles **840**, **845**, a 24 GHz mmWave band may be utilized for transmissions of data between vehicles **840**, **845**. For example, vehicles **840**, **845** may share real-time directional and navigation data with each other over the connection while the vehicles **840**, **845** pass each other across a narrow intersection line. Each vehicle **840**, **845** may be tracking the intersection line and providing image data to an image processing algorithm to facilitate autonomous navigation of each vehicle while each travels along the intersection line. In some examples, this real-time data may also be substantially simultaneously shared over an exclusive, licensed spectrum connection between the communication device **820** and the vehicle **845**, for example, for processing of image data received at both vehicle **845** and vehicle **840**, as transmitted by the vehicle **840** to vehicle **845** over the 24 GHz mmWave band. While shown as automobiles in FIG. 8, other vehicles may be used including, but not limited to, aircraft, spacecraft, balloons, blimps, dirigibles, trains, submarines, boats, ferries, cruise ships, helicopters, motorcycles, bicycles, drones, or combinations thereof.

(77) While described in the context of a 24 GHz mmWave band, it can be appreciated that connections may be formed in the system **800** in other mmWave bands or other frequency bands, such as 28 GHz, 37 GHz, 38 GHz, 39 GHz, which may be licensed or unlicensed bands. In some cases, vehicles **840**, **845** may share the frequency band that they are communicating on with other

vehicles in a different network. For example, a fleet of vehicles may pass vehicle **840** and, temporarily, share the 24 GHz mmWave band to form connections among that fleet, in addition to the 24 GHz mmWave connection between vehicles **840**, **845**. As another example, communication device **820** may substantially simultaneously maintain a 700 MHz connection with the mobile device **815** operated by a user (e.g., a pedestrian walking along the street) to provide information regarding a location of the user to the vehicle **845** over the 5.9 GHz band. In providing such information, communication device **820** may leverage antenna diversity schemes as part of a massive MIMO framework to facilitate time-sensitive, separate connections with both the mobile device **815** and the vehicle **845**. A massive MIMO framework may involve a transmitting and/or receiving devices with a large number of antennas (e.g., 12, 20, 64, 128, etc.), which may facilitate precise beamforming or spatial diversity unattainable with devices operating with fewer antennas according to legacy protocols (e.g., WiFi or LTE).

(78) The base station **810** and small cell **830** may wirelessly communicate with devices in the system **800** or other communication-capable devices in the system **800** having at the least a sensor wireless network, such as solar cells **837** that may operate on an active/sleep cycle, and/or one or more other sensor devices. The base station **810** may provide wireless communications coverage for devices that enter its coverages area, such as the mobile device **815** and the drone **817**. The small cell **830** may provide wireless communications coverage for devices that enter its coverage area, such as near the building that the small cell **830** is mounted upon, such as vehicle **845** and drone **817**.

(79) Generally, a small cell **830** may be referred to as a picocell and provide coverage for a local geographic region, for example, coverage of 200 meters or less in some examples. This may be contrasted with at macrocell, which may provide coverage over a wide or large area on the order of several square miles or kilometers. In some examples, a small cell **830** may be deployed (e.g., mounted on a building) within some coverage areas of a base station **810** (e.g., a macrocell) where wireless communications traffic may be dense according to a traffic analysis of that coverage area. For example, a small cell **830** may be deployed on the building in FIG. **8** in the coverage area of the base station **810** if the base station **810** generally receives and/or transmits a higher amount of wireless communication transmissions than other coverage areas of that base station **810**. A base station **810** may be deployed in a geographic area to provide wireless coverage for portions of that geographic area. As wireless communications traffic becomes more dense, additional base stations **810** may be deployed in certain areas, which may alter the coverage area of an existing base station **810**, or other support stations may be deployed, such as a small cell **830**. Small cell **830** may be a femtocell, which may provide coverage for an area smaller than a picocell (e.g., 100 meters or less in some examples (e.g., one story of a building)).

(80) While base station **810** and small cell **830** may provide communication coverage for a portion of the geographical area surrounding their respective areas, both may change aspects of their coverage to facilitate faster wireless connections for certain devices. For example, the small cell **830** may primarily provide coverage for devices surrounding or in the building upon which the small cell **830** is mounted. However, the small cell **830** may also detect that a device has entered its coverage area and adjust its coverage area to facilitate a faster connection to that device.

(81) For example, a small cell **830** may support a massive MIMO connection with the drone **817**, which may also be referred to as an unmanned aerial vehicle (UAV), and, when the vehicle **845** enters its coverage area, the small cell **830** adjusts some antennas to point directionally in a direction of the vehicle **845**, rather than the drone **817**, to facilitate a massive MIMO connection with the vehicle, in addition to the drone **817**. In adjusting some of the antennas, the small cell **830** may not support as fast a connection to the drone **817**, as it had before the adjustment. However, the drone **817** may also request a connection with another device (e.g., base station **810**) in its coverage area that may facilitate a similar connection as described with reference to the small cell **830**, or a different (e.g., faster, more reliable) connection with the base station **810**. Accordingly, the system

800 may enhance existing communication links in providing additional connections to devices that may utilize or demand such links. For example, the small cell **830** may include a massive MIMO system that directionally augments a link to vehicle **845**, with antennas of the small cell directed to the vehicle **845** for a specific time period, rather than facilitating other connections (e.g., the small cell **830** connections to the base station **810**, drone **817**, or solar cells **837**). In some examples, drone **817** may serve as a movable or aerial base station.

(82) The wireless communications system **800** may include devices such as base station **810**, communication device **820**, and small cell **830** that may support several connections to devices in the system **800**. Such devices may operate in a hierarchical mode or an ad-hoc mode with other devices in the network of system **800**. While described in the context of a base station **810**, communication device **820**, and small cell **830**, it can be appreciated that other devices that can support several connections with devices in the network may be included in system **800**, including but not limited to: macrocells, femtocells, routers, satellites, and RFLD detectors.

(83) Certain details are set forth above to provide a sufficient understanding of described embodiments. However, it will be clear to one skilled in the art that embodiments may be practiced without various of these particular details. The description herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The terms “exemplary” and “example” as may be used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

(84) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(85) Techniques described herein may be used for various wireless communications systems, which may include multiple access cellular communication systems, and which may employ code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), or single carrier frequency division multiple access (SC-FDMA), or any a combination of such techniques. Some of these techniques have been adopted in or relate to standardized wireless communication protocols by organizations such as Third Generation Partnership Project (3GPP), Third Generation Partnership Project 2 (3GPP2) and IEEE. These wireless standards include Ultra Mobile Broadband (UMB), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), LTE-Advanced (LTE-A), LTE-A Pro, New Radio (NR), IEEE 802.11 (WiFi), and IEEE 802.16 (WiMAX), among others.

(86) The terms “5G” or “5G communications system” may refer to systems that operate according to standardized protocols developed or discussed after, for example, LTE Releases **13** or **14** or WiMAX 802.16e-2005 by their respective sponsoring organizations. The features described herein may be employed in systems configured according to other generations of wireless communication systems, including those configured according to the standards described above.

(87) The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A

general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable read only memory (EEPROM), or optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor.

(88) Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Combinations of the above are also included within the scope of computer-readable media.

(89) Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(90) Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(91) From the foregoing it will be appreciated that, although specific embodiments have been described herein for purposes of illustration, various modifications may be made while remaining within the scope of the claimed technology. The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. An apparatus comprising: a transmitter configured to transmit a first message via a first antenna during a time period on a carrier of a radio frequency bandwidth; a noise calculator configured to

generate an interference signal during the time period based on the first message, the noise calculator comprising a network of processing elements that includes bit manipulation units, multiplication units, and accumulation units, wherein the bit manipulation units are configured to forward the first message to the accumulation units, wherein the accumulation units are configured to generate intermediate processing results using a nonlinear vector set, and wherein the accumulation units are configured to sum the intermediate processing results to generate the interference signal; and a receiver configured to receive a first receive signal having a first pre-selected portion of a second message via a second antenna during a first pre-defined time slot of the time period on the carrier and to receive a second receive signal having a second pre-selected portion of the second message via a third antenna during a second pre-defined time slot of the time period on the carrier, wherein the receiver is further configured to decode the first receive signal and the second receive signal based on the interference signal to recover the second message, wherein the second message is generated by a first user equipment (UE) or base station and the first receive signal is received from a second UE and the second receive signal is received from a third UE both acting as relays for the first UE or base station.

2. The apparatus of claim 1, wherein the receiver is configured to receive the first receive signal via a first set of antennas including the second antenna during the first pre-defined time slot of the time period on the carrier and is configured to receive the second receive signal via a second set of antennas including the third antenna during the second pre-defined time slot of the time period.

3. The apparatus of claim 1, wherein the receiver is configured to decode the first receive signal to retrieve the first pre-selected portion of the second message and to decode the second receive signal to retrieve the second pre-selected portion of the second message.

4. The apparatus of claim 1, wherein the receiver is configured to decode the first receive signal based on a portion of the interference signal generated during the first pre-defined time slot of the time period and to decode the second receive signal based on the second portion of the interference signal generated during the second pre-defined time slot of the time period.

5. The apparatus of claim 1, wherein the transmitter is configured to transmit the first message to a plurality of devices during the time period.

6. The apparatus of claim 1, wherein the first pre-defined time slot of the time period is non-overlapping with the second pre-defined time slot of the time period.

7. The apparatus of claim 1, wherein the receiver and the transmitter are each configured to implement one of a generalized selection combining protocol, a distributed space-time coding protocol, or an opportunistic relaying system protocol.

8. The apparatus of claim 1, wherein the transmitter is further configured to transmit a third message via a fourth antenna during the time period on the carrier of the radio frequency bandwidth, wherein the noise calculator is further configured to generate the interference signal during the time period further based on the third message.

9. The apparatus of claim 1, wherein the first pre-defined time slot and the second pre-defined time slot each comprise a symbol period, slot, or subframe according to a generalized selection combining (GSC), distributed space-time coding (DSTC), or opportunistic relaying selection (ORS) protocol.

10. An apparatus comprising: a first antenna configured to receive, via a carrier frequency, both of a first transmitted signal corresponding to a transmitted message and a first receive signal having a first message making up a first portion of a third message during a first pre-defined time slot of a time period; a second antenna configured to receive, via the carrier frequency, both of a second transmitted signal corresponding to the transmitted message and a second receive signal having a second message making up a second portion of the third message non-overlapping with the first message during a second pre-defined time slot of the time period, wherein the third message is generated by a first user equipment (UE) or base station and the first receive signal is received from a second UE and the second receive signal is received from a third UE both acting as relays for the

first UE or base station on the carrier frequency; a self-interference noise calculator configured to generate a first interference signal associated with the first antenna based on the transmitted message and to generate a second interference signal associated with the second antenna based on the transmitted message, wherein the self-interference noise calculator comprises a network of processing elements that includes bit manipulation units, multiplication units, and accumulation units, wherein the bit manipulation units are configured to forward the transmitted message to the accumulation units, wherein the accumulation units are configured to generate intermediate processing results using a nonlinear vector set, and wherein the accumulation units are configured to sum the intermediate processing results to generate the first interference signal; a first receiver configured to receive a first adjusted signal generated based on a combination of the transmitted signal, the first receive signal, and the first interference signal and to retrieve the first message based on the first adjusted signal; and a second receiver configured to receive a second adjusted signal generated based on a combination of the transmitted signal, the second receive signal, and the second interference signal and to retrieve the second message based on the second adjusted signal.

11. The apparatus of claim 10, further comprising: a first compensation circuit configured to combine the transmitted signal, the first receive signal, and the first interference signal to provide the first adjusted signal; and a second compensation circuit configured to combine the second signal, the second receive signal and the second interference signal to provide the second adjusted signal.

12. The apparatus of claim 10, further comprising a transmitter configured to transmit the transmitted message during a second time period that precedes the time period via a third antenna.

13. The apparatus of claim 10, wherein the first receiver and the second receiver are configured to reconstruct the third message based on a combination of the first message and the second message.

14. The apparatus of claim 10, wherein the first receiver and the second receiver are each configured to implement one of a generalized selection combining protocol, a distributed space-time coding protocol, or an opportunistic relaying system protocol.

15. A user equipment comprising: a receiver configured to, contemporaneous with transmission of a transmit signal on a carrier frequency, receive a first receive signal having a first pre-selected portion of a message on the carrier frequency via a first antenna during a first pre-defined time slot of a time period and a second receive signal having a second pre-selected portion of the message on the carrier frequency via a second antenna during a second pre-defined time slot of the time period, wherein the receiver is further configured to decode the first receive signal and the second receive signal to recover the message, wherein the message is generated by a first user equipment (UE) or base station and the first receive signal is received from a second UE and the second receive signal is received from a third UE both acting as relays for the first UE or base station on the carrier frequency; a transmitter configured to transmit the transmit signal on the carrier frequency; and a noise calculator comprising a network of processing elements that includes bit manipulation units, multiplication units, and accumulation units, wherein the bit manipulation units are configured to forward the transmit signal to the accumulation units, wherein the accumulation units are configured to generate intermediate processing results using a nonlinear vector set, and wherein the accumulation units are configured to sum the intermediate processing results to generate an interference signal used to decode the first receive signal and the second receive signal.

16. The user equipment of claim 15, wherein the receiver is configured to implement one of a generalized selection combining protocol, a distributed space-time coding protocol, or an opportunistic relaying system protocol.

17. The user equipment of claim 15, wherein the first pre-defined time slot of the time period is non-overlapping with the second pre-defined time slot of the time period.
